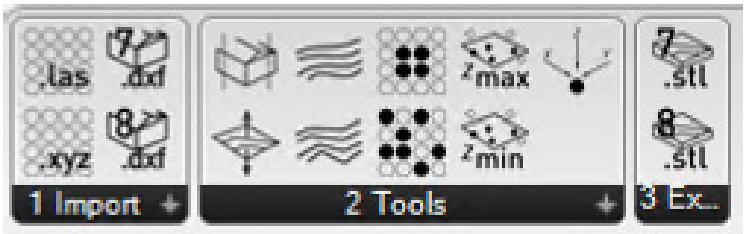
Murmeli Plugin

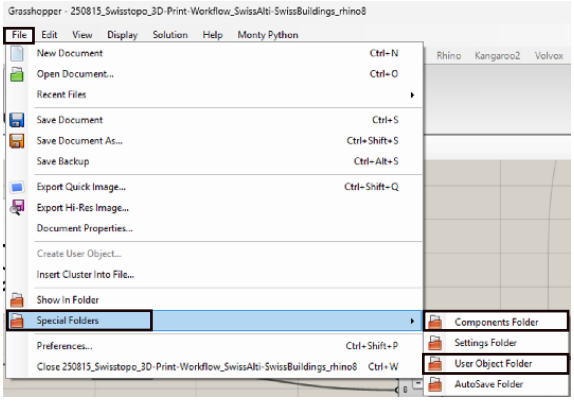
(Rhino 7 & 8)

Developped at ETH Zurich
LVML & Chair for Lanscape Architecture Prof. Voser
by Dennis Häusler, Matthias Vollmer, Nathan Lorethan

Licnece: MIT



The Plugin is structured in: 1 Import, 2 Tools, 3 Exports. If not specified the Operations work in Rhino 7 and 8. If they are exclusiv to one version they are marked with eighter 7 or 8. In Rhino 8, Ironpython is used. The three costum importers are tailered to handel specific data formats from swisstopo (.xyz / .las / .dxf). They might also work with similar data-sets from other source but have not been tested outside of swisstopo file-formats. The tools are costum operations for handeling the input data. The exporter allow for .stl-file exports for 3d printing in Rhino 7 & 8.



Installation

1 Copy & Past the Components into Grasshopper Special Folder/Components Folder

2 Copy & Past the User Objects into Grasshopper Special Folder/User Objects Folder

3 Close and Re-Open Grasshopper and Rhino

.las Importer

Imports one or more .las point cloud file with the option to randomly filter number of points. Default value is at 0.1% to prevent overload due to computers capabilities. The input pointcloud can also be filtered by Classification, default: all classes.

.xyz Importer

Imports one or more .xyz point cloud or raster file with the option to randomly filter the number of points. Mulitple lines in the .las-file can be skipped to avoid errors due to headers.

.dxf Importer

Imports a .dxf vector file and outputs a collection of meshes. For Rhino 7 no global shift is applied and for Rhino 8 a global shift from the Swiss Coordinate System LV95 towards 0/0/0 is applied and output aswell.

Advanced Mesh Clipper
Clips a mesh with a second mesh (clipper) and outputs a closed mesh. Comparable to the mesh intersection node, but functioning even with invalid or open mesh inputs.

Advanced Mesh Flipper
Takes a mesh and flips the normals if they are positive. So the result is a downwards facing mesh independent of the orientation of the input mesh.

Contours Simple
Creates simple contour lines on a Mesh. The height steps between the lines can be defined.

Contours Refined
Creates refined contour lines on a Mesh. The height steps between the lines can be defined with the option to define a smoothening process. Smooth, rounds the edges of the coutour lines while „Simplify" is reducing the number of segments on the countour lines.

Point Combiner
Combines up to four sets of points into one.

Point Thinner
Reduces the number of Points in a Set of Points randomly by eighter a given percentage or a given maximal Number.

Maximal z-Value
Finds the maximal z-Value among a set of points inside a given curve. Points on the curve are also concidered.

Minimal z-Value
Finds the minimal z-Value among a set of points inside a given curve. Points on the curve are also concidered.

Shift
Finds the center of a Point Set and outputs the vector from this found center to 0/0/0. Combining it with a move node the original set of point is shifted to and centered on 0/0/0.

.stl Exporter
Exports an .stl-file for 3D-printing when „export" is set to „true". Directory and Filename can be choosen. Two different Components have to be used for Rhino 7 and 8 as indicated.